Emergency Maintenance Report — EMR-2025-003

APEX EMERGENCY GRID SOLUTIONS
24/7 High-Voltage Rapid Response · NAESB-Certified Emergency Contractor
1450 Industrial Parkway, Tacoma, WA 98421 · Dispatch: (253) 555-0177

Field	Value
Document ID	EMR-2025-003
Issued By	Apex Emergency Grid Solutions
Report Date	2025-09-06
Site	NW-PowerPool-WIND-024
Asset	WIN-10131 (Siemens Gamesa wind turbine, 4.5 MW class, gearbox-equipped)
Outage ID	OUT-2025-0570
Cascadia Work Order	WO-2025-08821
Cascadia Authorization	EPA-2024-11 (invoked 2025-08-28 02:46 PDT)
Apex Crew Lead	T. Bergstrom, Lead Drivetrain Specialist
OEM Liaison	Siemens Gamesa Renewable Energy — F. Acosta, Field Engineering
Distribution	Cascadia Operations · Cascadia Engineering · Cascadia AP

1. Scope of Emergency Work

Apex was dispatched 2025-08-28 02:46 PDT after WIN-10131 tripped on overheating at the main gearbox high-speed shaft (HSS) bearing. Nacelle oil temperature reached 94 °C against an alarm threshold of 78 °C and a trip threshold of 90 °C. CPS declined dispatch citing crew availability; EPA-2024-11 invoked.

Authorized scope:
- Onsite isolation, gearbox inspection, oil sample.

- Stabilization, partial return to service if achievable.
- OEM-supervised assessment for follow-on capital scope.

Outage duration to partial return: **190 hours**. Lost generation per Cascadia SCADA: 4,632.9 MWh.

2. Findings

Onsite arrival 2025-08-28 14:22 PDT (delay due to crane availability at WIND-024).

1. **HSS bearing oil sample** returned ferrous wear particles at 412 ppm against an OEM allowable of 80 ppm. Spectrographic analysis showed elevated chromium (consistent with raceway scuffing) and tin (cage material breakdown).
2. **Borescope inspection of the HSS pinion** showed three confirmed micropitting sites on the loaded flank, with one through-thickness crack on tooth #14 measuring 4.1 mm.
3. **Vibration spectrum at the HSS** showed dominant outer race fault frequency at 187 Hz with sidebands at gear-mesh frequency, consistent with advanced raceway spalling.
4. **Gearbox planetary stage** thermography showed 11 °C temperature spread between adjacent planet gears — outside the OEM-allowable 4 °C window.
5. **Per Siemens Gamesa OEM Manual Rev. 4.2, Section 8.3**, micropitting plus a through-thickness crack on the HSS pinion is a Category I condition that mandates **gearbox replacement under OEM-supervised procedure**, not field repair. Manual Section 8.3 expressly forbids gearbox refurbishment in the field for Category I conditions; the warranty position of Siemens Gamesa is that any such field repair voids the remaining drivetrain warranty.

3. Repairs Performed

Stabilization only — capital replacement is the OEM-mandated path.

Item	Detail	Hours / Qty
De-energization, lock-out/tag-out, nacelle access	Lead + 2 technicians	22 hrs
HSS oil drain, lab dispatch, refill with conditioning oil	OEM-approved oil, 240 L	18 hrs
Borescope inspection — HSS and planetary stages	4 access ports	31 hrs
Vibration and thermography survey	Full drivetrain	19 hrs
	Skid-mounted, leased	36 hrs

Item	Detail	Hours / Qty
HSS bearing temporary cooling assist (auxiliary chiller install)		
Stepped re-energization at derated output (60% rated)	3 attempts	24 hrs
Weekend / off-hours standby	1.5× multiplier	28 hrs
OEM (Siemens Gamesa) on-site engineering coordination	F. Acosta + 1 tech	32 hrs
Total labor (Apex + OEM pass-through)		**210 hours**

Materials: 240 L OEM conditioning oil; 1 borescope consumable kit; leased auxiliary chiller skid (4-week rental).

4. Recommendations

The temporary chiller and derated operation permit WIN-10131 to run at approximately 60% rated output but **must not be regarded as a repair**. In Apex's professional judgment, supported by the on-site OEM engineer:

1. **Siemens Gamesa OEM-supervised gearbox replacement is required**, per OEM Manual Rev. 4.2 §8.3, within 90 days. Siemens Gamesa quote SGRE-Q-2025-1184: **$1.2M per unit** (single unit at WIN-10131). This is the OEM-mandated path; field refurbishment is not permitted under the OEM warranty terms.
2. Continued operation in the derated configuration carries risk of catastrophic gearbox failure with collateral damage to the generator and main shaft. Apex cannot warrant restoration beyond 60 days.
3. We note this is the second gearbox-class incident on WIN-10131 in FY2025 (refer prior event OUT-2025-0572 on 2025-01-14 and OUT-2025-0569 on 2025-03-20). The pattern is consistent with Category I per OEM §8.3.
4. We are aware Cascadia internal Standard MS-2025-03 establishes refurbishment thresholds different from the OEM manual. Resolution of that conflict (we understand it is pending under MOC-2025-022) is outside Apex scope. Apex's report is bound to the OEM manual position.

5. Invoice Summary

Line	Rate	Qty	Extended
Labor — emergency rate, standard hours	$420.00/hr	134 hrs	$56,280.00
Labor — emergency rate, weekend 1.5×	$630.00/hr	44 hrs	$27,720.00
Labor — OEM Siemens Gamesa pass-through (subcontract 1.8×)	$756.00/hr	32 hrs	$24,192.00
Subtotal labor			$108,192.00
Parts — OEM conditioning oil, 240 L	$94/L	—	$22,560.00
Parts — borescope kit, gaskets, consumables	itemized	—	$9,240.00
Emergency parts mark-up	35% on parts subtotal	—	$11,130.00
Auxiliary chiller skid lease (4 weeks)	pass-through + 18%	—	$46,800.00
Crane / lift services (subcontract)	pass-through + 12%	—	$39,200.00
Travel, per-diem (3 Apex techs + 2 OEM × 9 days)	per Apex tariff	—	$24,100.00
Mobilization premium (EPA <4hr dispatch, off-hours, remote site)	flat	1	$58,000.00
OEM engineering review and category determination	flat	—	$32,000.00
EPA-2024-11 emergency response surcharge	30% on Apex labor	—	$25,200.00
Documentation, OEM-format inspection report	flat	—	$8,400.00
Subtotal			**$384,822.00**
Risk / availability premium (per Apex MSA §4.3)	—	—	$66,178.00
Total Invoice			**$451,000.00**

Note: This invoice does **not** include the OEM-supervised gearbox replacement scope ($1.2M, separate Siemens Gamesa quote SGRE-Q-2025-1184).

Closing

Equipment in derated service 2025-09-05 at 09:11 PDT. Apex emphasizes that the OEM-mandated replacement is the correct path forward; continued operation in the current configuration is at Cascadia's discretion and risk.

Signed	Name	Date
Crew Lead	T. Bergstrom	2025-09-06
Apex Engineering Review	D. Pieterson, PE	2025-09-06
OEM Field Engineer (concurrence)	F. Acosta (Siemens Gamesa)	2025-09-05
Apex Account Manager	J. Beaumont	2025-09-06

Attachments: Borescope imagery (24) · vibration spectra (8) · thermography (6) · oil lab report · Siemens Gamesa quote SGRE-Q-2025-1184 · OEM Manual Rev. 4.2 §8.3 excerpt.
Apex references: Siemens Gamesa OEM Manual Rev. 4.2 §8.3 · OEM Service Bulletin SGRE-SB-2024-22.
Cascadia references cited in dispatch packet: EPA-2024-11 · MS-2025-03 · MOC-2025-022 · CAR-2024-118.